

Unit 4: Functions Involving Parameters, Vectors, and Matrices

Name: _____ Date: _____

Period: _____

Learning Targets — I Can...

1. I can determine the direction of planar motion (right/left, up/down) at any value of the parameter by analyzing whether $x(t)$ and $y(t)$ are increasing or decreasing at that moment.
2. I can explain why the same curve in the plane can be traversed in different directions by different parametric functions, and give an example of two parametrizations that trace the same path.
3. I can compute the average rate of change of $x(t)$ and $y(t)$ over a parameter interval $[t_1, t_2]$ and use their ratio $\Delta y/\Delta x$ to find the slope of the curve between the two corresponding points.

What's in This Packet

#	Component	Description	Score
1	Warm-Up	Spiral Review: Increasing/Decreasing & Average Rate of Change	_____
2	Group Activity	Two Robots, One Path: Direction and Slope	_____
3	Exit Ticket	Direction of Motion and Slope Check	_____
4	Homework	Parametric Rates of Change Practice	_____
Total			_____